Tiancong (Lindsay) Li

Research Interests: LLM Agents · Tool Learning · Efficient Language Models

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Education

University of California, Irvine **Sep. 2025 – Dec. 2026**

M.S. in Embedded and Cyber-Physical Systems | GPA: 3.78/4.00 *Irvine, CA*

- **Coursework:** AI Accelerator Design (in progress), Embedded Systems Modeling and Design, Real-Time Systems, Internet of Things Systems and Software, Wireless Sensor and Actuator Networks

University of California, Irvine, Division of Continuing Education **Sep. 2024 – Jun. 2025**

Certificate Program in Electrical Engineering and Computer Science | GPA: 3.93/4.00 *Irvine, CA*

- **Coursework:** Deep Learning, Computer Vision

Dalian University of Technology **Sep. 2021 – Jun. 2025**

B.Eng. in Software Engineering | Average: 85/100 *Dalian, China*

- **CS Coursework:** Machine Learning, Introduction to Artificial Intelligence, Data Structures and Algorithms
- **Math Coursework:** Probability and Statistics, Linear Algebra, Calculus, Discrete Mathematics

Publications

Tiancong Li, Wenzhuo Zhou. *ToolSIC: Small-Backbone Tool-Token Agents Rivaling Large Models.* Under review at ACL Rolling Review (ARR), October 2026.

Research Experience

ToolSIC: Parameter-Efficient Tool Agents with Tool Tokens **Oct. 2025 – Present**

First Author | Advisor: Prof. Wenzhuo Zhou, Department of Statistics, UC Irvine *Irvine, CA*

- Proposed **ToolSIC**, a tool-token agent that splits the agent loop at the tool-token interface into three lightweight modules (selection, invocation, composition) that share one transcript, with only one active at a time.
- Built from 0.6B modules (each 13× smaller than the released ToolGen-8B, 4.4× fewer parameters in total), the agent **outperforms ToolGen-8B by 10.9 points of grounded success** (41.2 vs. 30.3) and selects a correct tool more often (94.5% vs. 86.9%), at about 1/6 of its training compute and 1/11 of its inference compute.
- Found that trajectory training **suppresses tool tokens the trajectories never call** (top-1 selection 89%→12% at 0.6B), and avoided it by design: the selection module is the retrieval-stage checkpoint and never enters trajectory training.
- Showed that answer composition needs no training: an off-the-shelf instruction-tuned model beats every composition module fine-tuned on the benchmark's answers, of any size (+12.7 grounded success at 0.6B).
- Built a paired evaluation protocol (McNemar tests, seed noise floor, three judges, blind human comparison) and a **grounded-success** metric that exposes "hollow" judged successes, e.g., 52% of a BM25 retrieval pipeline's.

Selected Projects

Imitation Learning and VLA Policies for Robot Arms | *MECPS Capstone (Synaptics)* **Mar. 2026 – Present**

- Led software development and model training in a team of 4, building an imitation-learning pipeline for an SO-101 arm on Hugging Face LeRobot: data collection, training and real-robot evaluation code.
- Ongoing: **quantizing and deploying the trained policy on the Synaptics SL2619 edge NPU**, driven by natural-language commands.
- Trained and benchmarked four policies (ACT, SmolVLA, Diffusion Policy, Pi0.5) on real-robot pick-and-place; ACT succeeded on 10/10 trials, and an ablation showed that adding a second camera view and more demonstrations (50→90 episodes) raised success from 20% to 100%.

Intelligent Traffic Detection Drone | *Dalian University of Technology* **Apr. 2022 – Apr. 2023**

- Built a Raspberry Pi + STM32 drone combining deep-learning image recognition with sensor fusion for real-time traffic event detection; added a Blockly visual-programming interface. **3rd Prize, National Computer Design Competition**.

Technical Skills

ML & LLMs: PyTorch, Hugging Face Transformers, DeepSpeed, LoRA, constrained decoding, LLM-as-judge evaluation
Programming: Python, C, C++, MATLAB, Linux, Git
Systems: ARM Cortex-M, STM32, ESP32, Raspberry Pi, FreeRTOS, TensorFlow Lite (edge inference)